PREPAI: An LLM-Powered Framework for Automated Resume Analysis and Skill-Gap Identification

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Abstract

Students preparing for placements often face difficulty in determining how well their resumes align with a target job and identifying the skills they need to improve before an interview. Traditional resume screening and keyword-based approaches may overlook the contextual relevance between a student's qualifications and job requirements while providing limited personalized guidance for career preparation. To address these limitations, this project proposes PrepAI, an artificial intelligence-powered student career and interview preparation platform that combines resume–job matching with personalized skill-gap analysis and interview preparation. The framework consists of four major phases: (1) a resume and job-description processing module that extracts, cleans, and structures information from unstructured resume documents and target job descriptions; (2) a resume–job matching module that establishes a baseline using Term Frequency–Inverse Document Frequency, domain-specific weighting, and cosine similarity inspired by the selected base research, and further explores pre-trained semantic embeddings for improved contextual matching; (3) a student intelligence module that compares the student's competencies with the requirements of the target role to identify matched and missing skills; and (4) an artificial intelligence-powered preparation module that uses a Large Language Model to generate personalized technical and behavioural interview questions, preparation plans, and role-specific resume improvement suggestions. The system transforms unstructured resume information into actionable and explainable insights, enabling students to understand their suitability for a target role, recognize skill deficiencies, and prepare more effectively for placement opportunities. The complete framework is implemented as a web-based application using MongoDB, Express, React, and Node.js, integrating resume analysis, job matching, skill-gap identification, and personalized interview preparation into a unified platform.

References:

[1] M.-H. Ajjam and H. S. Al-Raweshidy, "AI-driven semantic similarity-based job matching framework for recruitment systems," Information Sciences, vol. 724, 2026, doi: 10.1016/j.ins.2025.122728.

Signature of the Guide

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